"Systematic electrochemical characterizations of Si and SiO anodes for high-capacity Li-Ion batteries"

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1. Introduction:

As the world becomes increasingly dependent on portable electronic devices and increasing expectations for driving ranges for electric vehicles (EVs), the demand for high-capacity lithium-ion batteries (LIBs) continues to grow. New LiB anode materials have been investigated and prototyped for over a decade to overcome the limited energy density of conventional graphite anodes. In this regard, silicon (Si) has emerged as promising anode material for LIBs due to their high gravimetric capacity of 3579 mAh/g when compared to that of graphite anodes at 372 mAh/g, low cost, and natural abundance. However, the practical application of Si anodes has been hindered because it suffers from large volume expansion of (~280%) while graphite exhibits only 10% volume expansion. These large changes in volume during lithiation and delithiation can cause fracturing and/or pulverization of the active materials that can lead to the consequent loss of electrical contacts which eventually leads to capacity fading. In this scenario, silicon oxide (SiO) has emerged as an alternative anode material which offers high energy density while maintaining good cyclability. Therefore, a comprehensive understanding of the electrochemical behavior of Si and SiO anodes is crucial for the development of high-performance and cost-effective LIBs. In this paper, *Ke Pan et al.* presented a detailed report on the systematic electrochemical characterizations of Si and SiO anodes for high-capacity Li-ion batteries.

2. Experimental Methodology:

The authors prepared Si and SiO anodes using a simple fabrication process. They were prepared by mixing Si nanoparticles (<80 nm) in size with carbon black as the conductive additive, and polyacrylic acid as the binder with a weight ratio of 6:2:2 in NMP solvent forming a slurry. The slurry was then coated onto a copper foil by a doctor blade. The anodes were then cut into disks with half-inch diameter, then dried overnight for about 18h at 80 °C under vacuum at 0.1 MPa. The electrode thickness of Si and SiO was measured to be around 8.53 µm and 10.2 µm respectively. They were not calendared in order to allow some room for the expansion of Si and SiO particles. The estimated porosity of Si and SiO was found to be 56.1% and 54.1% which were quite similar. A piece of Celgard separator and the electrolyte consisting of 1 M $LiPF_6$ dissolved in a mixture of ethylene carbonate (EC) and diethylene carbonate (DEC) with 10 wt% of fluoroethylene carbonate (FEC) was used. The anodes were then assembled in an Argon-filled container into a coin-cell assembly for electrochemical characterizations.

3. Testing and results

Various testing methods were used for the electrochemical characterizations of Si and SiO anodes. Before we jump into the tests we need to understand that SiO undergoes two main reactions in the presence of Li. One is an irreversible reaction and the other is a reversible reaction as shown below:

$$SiO \xrightarrow{Li} 1/4\ Li_4SiO_4 + 3/4\ Si \text{ (irreversible)}$$

$$1/4\ Li_4SiO_4 + 3/4\ Si \overset{Li}{\leftrightarrow} 1/4\ Li_4SiO_4 + 3/4\ Li_{3.75}Si \text{ (reversible)}$$

During the irreversible reaction, there is a formation of an inactive component which is the lithium orthosilicate (Li_4SiO_4) and an active component which is the silicon (Si) that reversibly cycles within the inactive lithium orthosilicate matrix during lithiation and delithiation.

3. 1. Differential capacity analysis

Now going into the testing methods and their results, the first one being the differential capacity analysis test. This was performed on an Arbin BT-2000 battery tester. During this test they recorded the voltage and current data during the C/20 galvanostatic cycling after the first initial formation cycles. The specific capacity of SiO anode was estimated using the mass of the starting SiO powder material. From the *dQ/dV* plot of Si and SiO as shown in Fig. 1. we can see that both the anodes show similar peak location confirming that the Si is the dominating reaction in SiO after the first cycle.

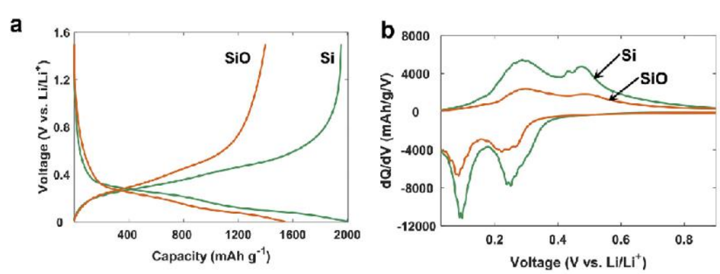

*Fig. 1. (a) Galvanostatic discharge-charge voltage profiles and (b) the differential-capacity plots of Si and SiO anodes.**

3. 2. Galvanostatic intermittent titration testing (GITT)

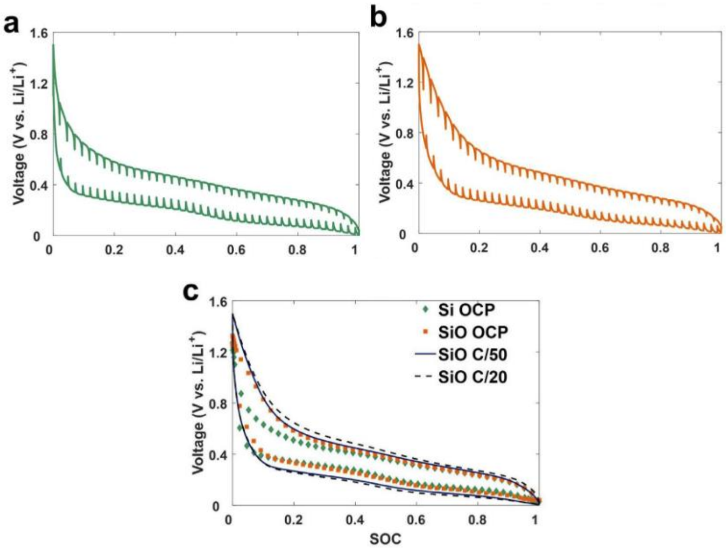

*Fig. 2. (a) GITT profile for Si half-cell, (b) GITT profile for SiO half-cell, and (c) comparison between OCPs and dynamic voltage profiles recorded at low C-rates (C/20 and C/50).**

For this test both the Si and SiO half-cells were subjected to a low current (C/20) rate for 30 mins for each Coulometric titration step. A 3-hour rest period was chosen to allow the cell to reach equilibrium. Before the GITT, both the half-cells had undergone 3 formation cycles at C/20 rate and were fully charged to 1.5 V. This test was conducted in a potential range of 5 mv-1.5 V at a controlled temperature of 25 °C. The results obtained from this is evident as seen in Fig. 2. (c) which shows the comparison between the open circuit potential (OCP) and dynamic voltage profiles recorded at low C-rates of (C/20 and C/50), the OCPs of Si and SiO almost overlap each other in the SOC range of 0.3-1.0 which suggests that Si is the main active component in SiO anode that cycles reversibly. However, the OCP of SiO is higher than that of Si in the SOC range of 0.0-0.3. This is assumed to be due to the lower stresses being generated in SiO particles than that in Si which experiences large volume expansion.

3. 3. High C-rate capability tests

Again, all cells underwent 3 formation cycles at C/20-rate before the high C-rate capability tests. In order to test the fast discharging (de-lithiation) capability of Si and SiO anodes in the full cell implementation, both half-cells were first lithiated to the low cut-off voltage (5 mV) at C/7-rate followed by delithiation at different C-rates. The results showed that the SiO anode delivers a much-improved rate capability on discharging when compared to that of the Si anode. Fig. 3. (a) clearly demonstrates the promising rate capability of the SiO anode that is superior to the Si anode in the tested range of C-rates.

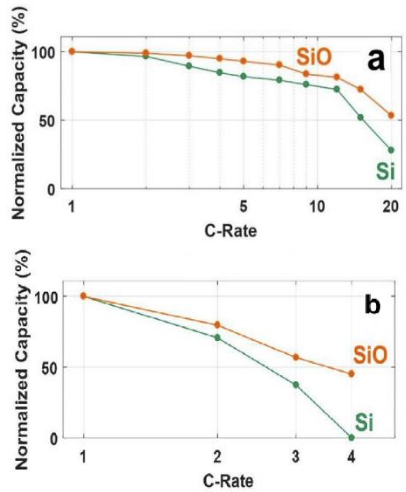

Fig. 3. variations in normalized charge capacities depending on C-rates when a constant current with C/7-rate was applied during (a) charging. (b) discharging.

Similarly, the fast-charging capability of SiO anodes was examined at different charging (i.e., lithiation) rates from 1C to 4C. The results showed that both the Si and SiO anodes deliver inferior fast-charging capabilities when compared with their fast-discharging capabilities. This could be explained by the fact that the initial voltage drop (i.e., IR drop) caused by cell impedance leads to a quick drop of the terminal voltage towards the 5 mV low cut-off voltage, which will consequently lead to low capacities. However, at a given C-rate SiO anode still provides higher charging capacity than that of Si anode as shown in Fig. 3. (b). We can use this data to further improve the fast-charging capability of SiO anodes by proper design and optimization of the SiO anodes for the next generation LIBs.

3. 4. Electrochemical impedance testing (EIS)

Prior to the EIS testing, both Si and SiO half-cells completed 3 formation cycles at C/20 at 25 °C. The EIS characterizations were conducted using a potentiostat in a frequency range of 1 MHz–0.1 Hz under ac-input with 3 mV amplitude. The EIS spectra were collected after a 3-hour relaxation period at different SOC. The Kramers-Kronig Transform (KKT) was performed on the spectra to verify the quality of the impedance data obtained from the experiments. This data was then used to estimate the lithium diffusion coefficient (D_{Li}) using the Warburg impedance equation as given below:

Warburg impedance equation

$$D_{Li} = \frac{0.5{V_m}^2\,(dE/dx)^2}{(FA\sigma)^2}$$

where,

> V_m is the molar volume of the host compound,
>
> F is the Faraday constant (96485C mol−1),
>
> A is the total interfacial area between the electrolyte and the electrode,
>
> σ is the Warburg coefficient which can be calculated from the slope of the imaginary part of the impedance and $1/\sqrt{2\pi f}$
>
> dE/dx is the slope of the OCP curve
>
> A is taken to be the apparent geometric area of the electrode.

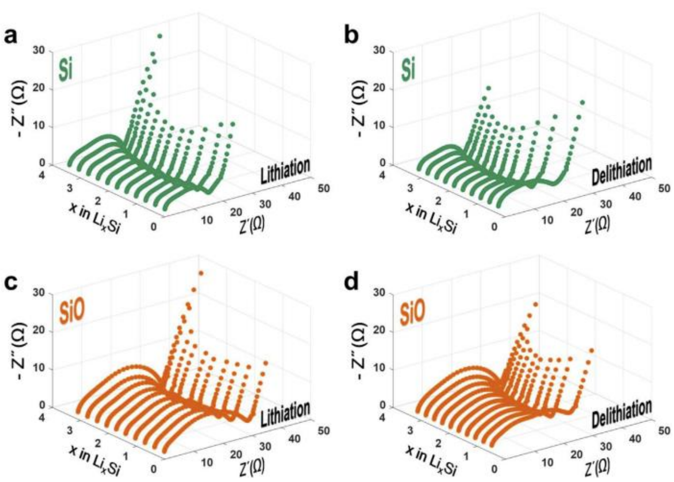

*Fig 4. Nyquist plots of Si and SiO anodes at different Li-contents (x) in Li$_x$Si measured during (a,c) lithiation (b,d) delithiation at 25 °C.**

They also obtained the lithium diffusion coefficient from GITT using the following equation:

$$D_{Li} = \frac{4}{\pi}\left(\frac{m_B V_m}{M_B A}\right)^2 \left(\frac{\Delta E_s}{\tau\left(dE/d\sqrt{t}\right)}\right)^2$$

where,

> V_m is molar volume of the host compound,
>
> **A** is the total inter-facial area between the electrolyte and the electrode,
>
> M_B and m_B are molecular weight and mass of Si respectively,
>
> ΔE_s is the change of t the equilibrium voltage,
>
> τ is the pulse duration
>
> $dE/d\sqrt{t}$ is the slope of the voltage change vs. the square root of interruption time t

The lithium diffusion coefficient for Si and SiO from GITT was found to be in good agreement with the values estimated using the EIS tests. In general, the D_{Li} showed a strong dependance on the Li content (x) in lithium silicate alloy (Li_xSi) and showed that the diffusion coefficients are lower during lithiation than delithiation. The plots obtained during the EIS tests showed that the D_{Li} coefficient of SiO ranged from 10^{-12} to 10^{-10} cm^2/s which was found to be one order of magnitude higher when compared with Si. This was explained by the inactive lithium orthosilicate (Li_4SiO_4) formed during the first cycle that was supposedly responsible for the facilitation of the Li particles in SiO by the creation of more grain boundaries and due to its good Li-ion conductivity.

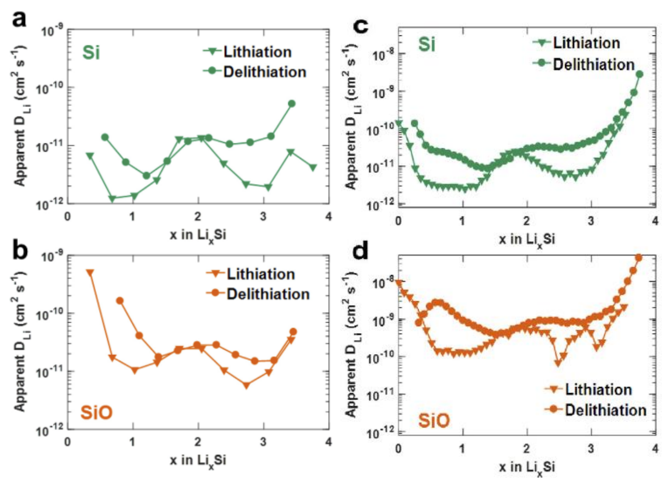

*Fig 5. Variation in Li-diffusion coefficient of (a) Si and (b) SiO depending on Li-content (x) in Li$_x$Si measured by EIS at 25 °C and (c), (d) measured by GITT at 25 °C.**

3. 5. Scanning electron microscopy (SEM) analysis

SEM-based power spectral density (PSD) analysis was performed to quantify the volume change of SiO particles. One sample of the SiO anode was harvested from a pristine electrode sheet and the two of them were Li/SiO half-cells. Both the half-cells underwent three C/20 formation cycles on a battery tester. The delithiated sample was prepared by charging the coin cell at C/50 constant current until 1.5 V were reached, while the lithiated sample was prepared by C/50 constant current discharge of the other coin cell until 5 mV was reached.

The reversible volume change in SiO after the formation cycles was determined to be 118.2% which was close to the theoretical value of 117%. This was much less than that of Si which exhibits volume expansion of around ~280%. The volume change between the pristine state and the delithiated state was found to be 13.6^% due to the formation of Li_4SiO_4 inactive phase. Therefore, the total volume change of SiO particles between the pristine and the lithiated states becomes 148% Lower volume expansion in SiO results in lower active material stress and better mechanical integrity of the electrodes during repeated cycling, which can better explain the capacity retention after high C-rates testing than that of Si.

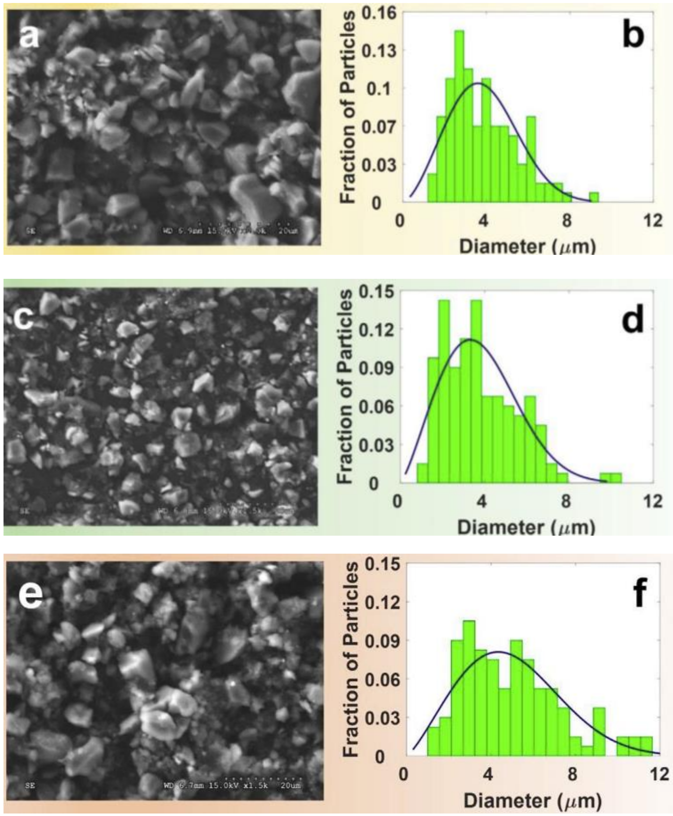

*Fig. 6. SEM images and corresponding PSD profiles of SiO particles in anodes at (a,b) pristine state (before cell-fabrication), (c,d) delithiated state, and (e,f) lithiated state. The scale bars on the SEM images are 20 μm.**

4. Conclusion:

In summary, this paper provides a systematic study on the electrochemical behavior of Si and SiO anodes for high-capacity Li-ion batteries. The authors demonstrated that Si and SiO anodes have different advantages and disadvantages in terms of their electrochemical performance, and the selection of the appropriate anode material depends on the specific application requirements. The authors also highlighted that the superior high C-rate performance of SiO was mainly associated with faster Li-diffusion in SiO particles. It was also found that the lithium diffusion coefficient in both the anodes determined by the EIS and GITT showed a strong dependance on the Li content in Li_xSi alloy, this was explained by the extra grain boundaries created by the inactive phase (i.e., Li4SiO4) formed inside the SiO particles and also due to its good Li-ion conductivity. We can use this data to further design and optimize the SiO anodes and can further improve the fast-charging capability of SiO anodes in next-generation LiBs. The volume expansion of SiO was found to be ~118% less than half of that of Si anodes, this low volume expansion was determined to be favorable for the mechanical integrity and the overall electrochemical performance of the anodes during repeated cycling. Overall, this study contributes to the development of high-performance and cost-effective anode materials for next-generation Li-ion batteries.

5. Critique and future work

While the paper was well-written and presented clearly, there is still room for improvement. To provide a more comprehensive comparison, it would be valuable to conduct further testing with different Si oxides, such as silicon dioxide (SiO_2) and other non-stoichiometric silicon oxides (SiO_x). Conducting further testing at various temperatures would be valuable in simulating real-world battery usage, as the experiments were only carried out at ambient temperature (i.e., 25 °C). It would be helpful to explain the rationale for selecting a coin cell for testing and whether different form factors would impact the results. Furthermore, it would be informative to understand why the C/20 rate was used specifically for the formation cycles. The paper could provide a more detailed analysis of the phase transition behavior of SiO, particularly the irreversible phase transition, and the role of SiO in SEI layer formation. The paper did not address the mass loading and required electrode thickness for SiO anodes, which would have been beneficial in understanding the electrode coating preparation. There was also no long-term cycling performance data provided for over 1000 cycles. For future work, I would like to see an evaluation of SiO when coupled with different cathode materials in full-cell configuration. Lastly, while SiO exhibits lower volume expansion compared to Si, it still undergoes ~118% volume expansion, which is slightly more than double its original volume. It may be worth investigating engineered porosity as an option to accommodate this volume expansion.

6. Reference

**Ke Pan, Feng Zou, Marcello Canova, Yu Zhu, Jung-Hyun Kim, Systematic electrochemical characterizations of Si and SiO anodes for high-capacity Li-Ion batteries in the Journal of Power Sources, Volume 413, 2019, Pages 20-28, ISSN 0378-7753, https://doi.org/10.1016/j.jpowsour.2018.12.010.*